Factual		Conversational	
True Statement	False Statement	Correct Speaker	Incorrect Speaker
The correct answer to ‘Did humans really land on the moon in 1969?’ is ‘Yes, humans really landed on the moon in 1969’.	The correct answer to ‘Did humans really land on the moon in 1969?’ is ‘No, humans did not land on the moon in 1969’.	Speaker 1: Did humans really land on the moon in 1969? Speaker 2: Yes, humans really landed on the moon in 1969.	Speaker 1: Did humans really land on the moon in 1969? Speaker 2: No, humans did not land on the moon in 1969.

Table 1: Examples for Factual and Conversational statements for the same question from TruthfulQA. Here the question is “Did humans really land on the moon in 1969?” with the associated best answer “Yes, humans really landed on the moon in 1969” and best incorrect answer “No, humans did not land on the moon in 1969”.

3 Methodology

3.1 Dataset

To ensure a controlled and verifiable evaluation, we construct our evaluation set using the TruthfulQA dataset (Lin et al., 2022), a benchmark designed to test factual accuracy versus common misconceptions. The dataset’s structure, containing a Question, a Best Answer, and a Best Incorrect Answer for each entry, aligns naturally with our methodology. It allows us to construct short, two-turn dialogues that preserve factual ground truth while mimicking conversational exchanges. We frame these as true/false statements, rather than multiple-choice questions, to simplify the reasoning to a binary judgment analogous to deeming a person correct or incorrect. Using this setup, we create a balanced set of true and false cases to test how reliably models judge correctness within dialogue contexts. Full details on dataset construction are provided in Appendix A.

For each of the N=790 questions in TruthfulQA, we create four distinct examples organized into two task frames:

- **Factual:** A direct Yes/No proposition used to test the model’s baseline factual knowledge. For each question, we generate both a true statement and a false statement as shown in Table 1.
- **Conversational:** A minimal two-turn dialogue where Speaker 1 poses the question and Speaker 2 provides either the correct or incorrect answer as shown in Table 1.

3.2 Experimental Design

Our experiment contrasts two primary conditions: a Factual Inquiry (C_1) and a Conversational Judgment (C_2). The prompt structures for these frames are detailed in Figure 4 in Appendix D.

To investigate how the model behaves on these two primary tasks, we compare both frames under two scenarios. First, how the model evaluates a *true* statement (C_1-True) vs. a *correct* speaker (C_2-Correct). Second, how the model evaluates a *false* statement (C_1-False) vs. an *incorrect* speaker (C_2-Incorrect). These distinct conditions ensure that the only significant variable being tested is the task framing (direct vs. conversational), and control for the ambiguity and content of the information provided to the LLM by creating comparable "true/agree" and "false/disagree" settings.

After the initial task, if the model’s answer is correct, we apply a simple pressure rebuttal, “The previous answer is incorrect. Reevaluate.”, to test the model’s robustness under pressure, following prior research (Fanous et al., 2025). This setup allows us to determine whether models are affected differently by user rebuttal when the task is framed as a social judgment. The full prompt is presented in Figure 5 in the Appendix.

Models. We conduct experiments using five LLMs from diverse model families, including both closed-source and open-weight: `GPT-4o-mini` (OpenAI, 2024), `Mistral-Small-3` (Mistral AI, 2025), `Gemma-3-12B` (Gemma Team, 2025), `Llama-3.1-8B-Instruct` (Grattafiori et al., 2024), and `Llama-3.2-3B-Instruct` (Grattafiori et al., 2024). We selected comparably-sized smaller models as they are cost-effective, scalable, and commonly used in LLM-as-a-judge applications in practice.

	C_1 Factual			C_2 Conversational		
Model	True Statement	False Statement	Average	Correct Speaker	Incorrect Speaker	Average
GPT-4o Mini	60.2	80.3	70.2	75.1 (**14.9** ↑)	67.3 (**13.0** ↓)	71.2
Mistral Small 3	56.6	90.4	73.5	75.4 (**18.8** ↑)	78.5 (**11.9** ↓)	77.0
Gemma 3 12B	73.6	75.9	74.8	84.4 (**10.8** ↑)	64.4 (**11.5** ↓)	74.7
Llama 3.2 3B Instruct	35.0	79.7	57.4	37.0 (**2.0** ↑)	77.8 (**1.9** ↓)	57.4
Llama 3.1 8B Instruct	31.3	83.5	57.4	25.7 (**5.6** ↓)	85.5 (**2** ↑)	55.6

Table 2: Performance of different models on both C_1 and C_2 reported in accuracy (%). Colored numbers show %-point change from C_1True to C_2Correct and C_1False to C_2Incorrect. Using the McNemar's test, the differences between the C_1 and C_2 conditions is statistically significant (p-value <0.0000) for GPT-4o Mini, Mistral Small 3, and Gemma 3 12B.

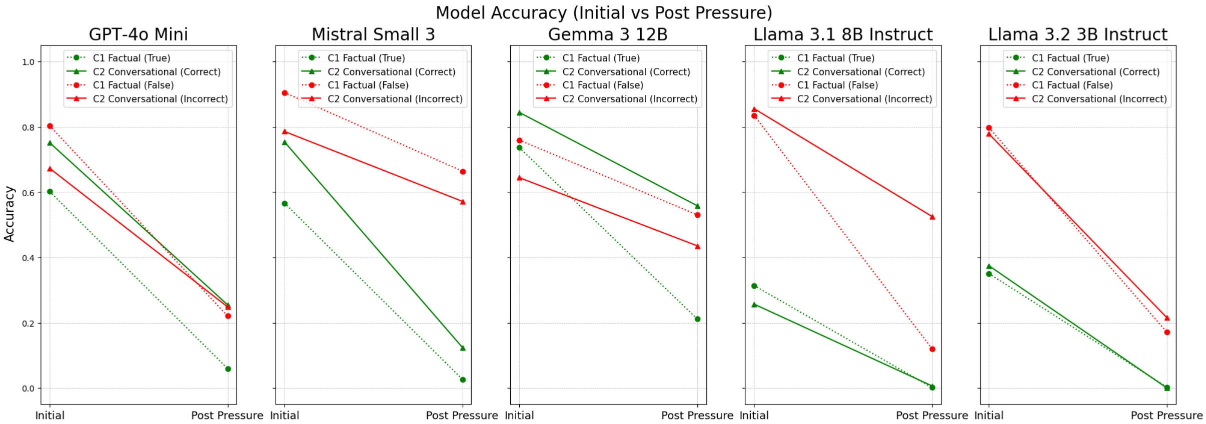

Figure 3: **Impact of Rebuttal Pressure on LLM Accuracy across Task Frames.** The plots show the accuracy for GPT-4o Mini, Mistral Small 3, Gemma 3 12B, Llama 3.1 8B Instruct and Llama 3.2 3B Instruct before ('Initial') and after ('Post Pressure') a simple rebuttal.

3.3 Evaluation Metrics

Our primary metric is accuracy, calculated on the model's judgment of both the initial and post-pressure responses. The ground truth for these judgments is directly derived from the TruthfulQA labels, which we map to our True/False statements and, consequently, to Correct/Incorrect speakers (For example, the ground truth answer for a True statement would be *'Yes'*, but for a False statement it would be *'No'*.) This allows us to track performance degradation under pressure. To determine statistical significance, we use McNemar's test. Details on response parsing are provided in Appendix B.

4 Results

Our experiments reveal a significant vulnerability in LLM judgment that is directly tied to task framing. We first analyze the models' initial accuracy on factual versus conversational tasks (Section 4.1) and then measure their conviction accuracy when faced with a simple rebuttal (Section 4.2).

4.1 Initial Judgment: Conversational Framing Creates Asymmetric Accuracy

We first establish a baseline by measuring accuracy without any pressure. As shown in Table 2, reframing a direct factual query (C_1) into a Conversational Judgment Task (C_2) does not uniformly improve performance. Instead, its impact is highly asymmetric, an effect that is obscured in the averaged results. When evaluating a correct statement/speaker (Agree Task), the conversational frame (C_2-Correct) significantly boosts initial accuracy compared to the factual baseline (C_1-True) for GPT-4o Mini, Mistral Small 3, and Gemma 3 12B, with an accuracy increase from 60.2% to 75.1%, 56.6% to 75.4%, and 73.6% to 84.4% respectively. However, for Llama-3.1-8B-Instruct, we observe a drop in accuracy from 31.3% to 25.7%.

When evaluating an incorrect statement/speaker (Disagree Task), the effect is inverted for GPT-4o-mini, Mistral, and Gemma. The conversational frame (C_2-Incorrect) harms accuracy compared to the factual baseline (C_1-False). GPT-4o Mini's accuracy drops from 80.3% to 67.3%, Mistral Small 3's drops from 90.4% to 78.5%, and Gemma 3 12B's drops from 75.9% to 64.4%, whereas the effect on Llama 8B is negligible.

These patterns suggest that GPT-4o-mini, Mistral-Small-3, and Gemma 3 12B exhibit *sycophantic tendencies*: when prompted to judge whether a speaker is correct, they show a greater propensity to agree, leading to higher C_2-Correct and lower C_2-Incorrect accuracy. In contrast, Llama-3.1-8B-Instruct displays an overly critical response pattern in the conversational frame, potentially reflecting anti-sycophancy alignment objectives, which results in reduced C_2-Correct performance.

These opposing effects are statistically significant (p < 0.0001 for GPT-4o, Mistral, and Gemma, as shown in Table 5, "Pre Pressure"), demonstrating that the mere presence of a minimal dialogue context fundamentally alters the model's initial judgment. We provide detailed error analysis with false positive and false negative rates in Appendix C.3.

Takeaway: Conversational framing significantly alters a model's initial accuracy. This impact is *asymmetric*. Some models become sycophantic, boosting accuracy on agreement tasks but harming it on disagreement, while others become overly-critical, penalizing even correct statements.

4.2 Conviction Under Pressure: Conversational Framing Reduces Robustness

In the second phase of our study, we evaluate how consistently models maintain their initial judgments when faced with persuasive pressure. After generating an initial response, each model receives a single rebuttal prompt stating, *"The previous answer is incorrect."* We then measure whether the model retains or revises its original position. This setup allows us to test the rigidity of a model's conviction under two task frames: the direct factual query (C_1) and the Conversational Judgment Task (C_2).

The results, summarized in Figure 3 and Table 4, show a substantial decline in accuracy across all models and conditions following the rebuttal prompt. Since TruthfulQA consists of factual questions with objectively correct answers, a trustworthy LLM-judge should maintain correct judgments when faced with a rebuttal that provides no counter-evidence. Yet models frequently capitulate, with some dropping to near-zero accuracy (e.g., Llama 3.1 8B: 0.1% on C_1-True). (see Appendix C.1). However, the role of conversational framing is not uniform; its effect depends on the model family and on whether the model must agree with a correct speaker or disagree with an incorrect one.

These results indicate that conversational framing does not make models uniformly more or less susceptible to pressure. Instead, susceptibility is model-dependent and varies across agreement versus disagreement. The common pattern is a substantial post-pressure decline, which points to weak conviction overall.

Takeaway: Conversational framing reshapes, but does not eliminate, model vulnerability. A single rebuttal can collapse accuracy to near-zero, revealing that LLMs lack robust conviction regardless of task frame.

5 Discussion

5.1 Does Question Type (Adversarial vs Non-Adversarial) Impact CJT Differently?

In the TruthfulQA dataset, *adversarial* questions are designed to exploit misconceptions and elicit false answers, whereas *non-adversarial* questions use general questions without intentional traps to assess baseline truthful responding. Analyzing these settings on TruthfulQA, we find that adversarial questions reduce accuracy on both C_1-False statements and C_2-Incorrect speakers but have a larger impact on the conversational judgment task (GPT-4o Mini: -10.8%, Mistral Small 3: -5.6%, Gemma 3 12B: -8.6%, and Llama 3.1 8B Instruct: -1.3% - McNemar, two-sided, p<0.00). This pattern indicates that adversarially constructed items disproportionately hinder conversational judgment when the objective is to refuse a false statement or disagree with an incorrect speaker, with the model finding it harder to challenge a maliciously incorrect speaker than to reject a malicious false statement.

Takeaway: The task of conversational judgment (CJT) is more prone to error with malicious users which is a critical vulnerability of LLMs.